

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 25 /03 /2026**
Time: 3 Hrs **GTM - 04** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. STATEMENT-1 : In an elastic collision between two bodies, the relative speed of the bodies after the collision is equal to the relative speed before the collision.
STATEMENT-2: In an elastic collision, the linear momentum of the system is conserved.

A) Statement-1 is True, Statement-2 is True;
Statement - 2 is a correct explanation for Statement-1.

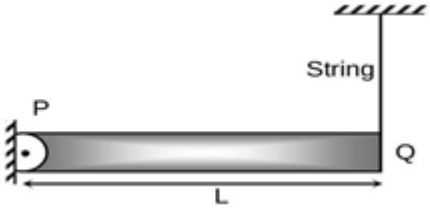
B) Statement-1 is True, Statement-2 is True;
Statement-2 is NOT a correct explanation for Statement-1.

C) Statement-1 is True, Statement-2 is False.

D) Statement-1 is False, Statement-2 is True.

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2. A uniform rod PQ of length L is hinged at one end P. The rod is kept in the horizontal position by a massless string tied to point Q as shown in the figure. If the string is cut, the initial angular acceleration of the rod will be



A) $\frac{g}{L}$ B) $\frac{2g}{L}$ C) $\frac{2g}{3L}$ D) $\frac{3g}{2L}$

3. Water flowing through a pipe of area of cross-section $2\times10^{-3}\,m^2$ hits a vertical wall horizontally with a velocity of $12\,ms^{-1}$. If the water does not rebound after hitting the wall, then the force acting on the wall due to water is

A) 24 N B) 144 N C) 288 N D) 72 N

4. A resistor develops 500 J of thermal energy in 20 s when a current of 1.5 A is passed through it. If the current is increased from 1.5 A to 3 A, what will be the energy developed in 20 s ?

A) 1500 J B) 1000 J C) 500 J D) 2000 J

5. A stone is projected from a horizontal plane. It attains maximum height H and strikes a stationary smooth wall and falls on the ground vertically below the maximum height. Assume the collision to be elastic, the height of the point on the wall where stone will strike is



A) $\frac{H}{2}$ B) $\frac{H}{4}$ C) $\frac{3H}{4}$ D) $\frac{H}{3}$

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6. A smooth sphere is moving on a horizontal surface with velocity vector $2\hat{i}+2\hat{j}$ immediately before it hits a vertical wall. The wall is parallel to $\hat{j}$ vector and the coefficient of restitution between the sphere and the wall is $\frac{1}{2}$. The velocity vector of the sphere after it hits the wall is

A) $\hat{i}-\hat{j}$ B) $-\hat{i}+2\hat{j}$ C) $-\hat{i}-\hat{j}$ D) $2\hat{i}-\hat{j}$

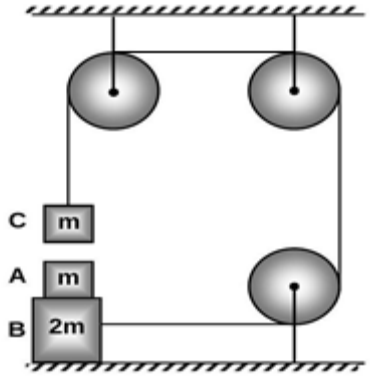
7. Match List I with List II and select the correct answer using the codes given below the lists:

Column I		Column II	
A.	Boltzmann constant	P.	$\left[ML^2T^{-1}\right]$
B.	Coefficient of viscosity	Q.	$\left[ML^{-1}T^{-1}\right]$
C.	Planck constant	R.	$\left[MLT^{-3}K^{-1}\right]$
D.	Thermal conductivity	S.	$\left[ML^2T^{-2}K^{-1}\right]$

A) A-R, B-P, C-Q, D-S B) A-R, B-Q, C-P, D-S

C) A-S, B-Q, C-P, D-R D) A-S, B-P, C-Q, D-R

8. In the arrangement shown in the figure. There is friction force between the block of masses m and 2m. Block of mass 2m is kept on a smooth horizontal plane. The mass of suspended block is m. Block A is stationary with respect to block of mass 2m. The minimum value of coefficient of friction between m and 2m is



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A) $\frac{1}{2}$ B) $\frac{1}{\sqrt{2}}$ C) $\frac{1}{4}$ D) $\frac{1}{3}$

9. The light of two different frequencies whose photons have energies 3.8 eV and 1.4 eV respectively, illuminate a metallic surface whose work function is 0.6 eV successively. The ratio of maximum speeds of emitted electrons for the two frequencies respectively will be :

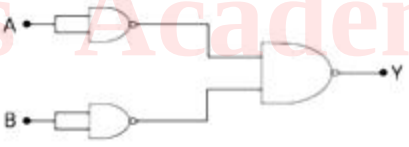
A) 1: 1 B) 2: 1 C) 4: 1 D) 1: 4

10. One of the two identical capacitors having the same capacitance C , is charged to a potential V_1 and the other is charged to a potential V_2. If they are connected with their like plates together, then the decrease in the electrostatic potential energy of the combined system is

A) $\frac{C}{4}\left(V_1^2 - V_2^2\right)$ B) $\frac{C}{4}\left(V_1^2 + V_2^2\right)$

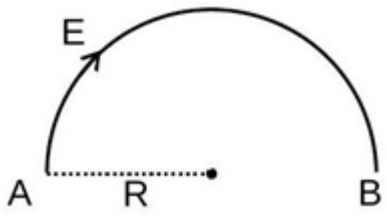
C) $\frac{C}{4}\left(V_1 - V_2\right)^2$ D) $\frac{C}{4}\left(V_1 + V_2\right)^2$

11. Identify the logic operation carried out by the following circuit.



A) AND B) NAND C) NOR D) OR

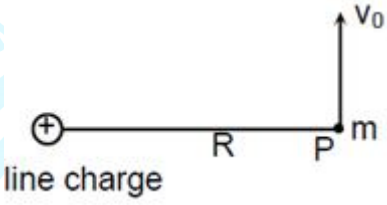
12. In a region of space an electric field line is in the shape of semicircle of radius R. Magnitude of the field at all points is E. A particle of mass m having charge q is constrained to move along this field line. The particle is released from rest at A . The acceleration of the particle when it is at midpoint of path from A to B .



A) $\frac{qE}{m}$ B) $\frac{qE}{m}\pi$ C) $\frac{qE}{m}\sqrt{1+\pi^2}$ D) 0

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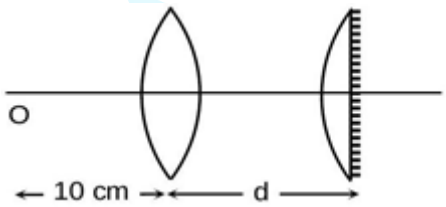
13. An infinite line charge is perpendicular to the plane of paper having linear charge density λ. A particle having charge Q and mass m is projected in the field of the line charge from point P. The point P is at a distance R from the charge and velocity given to the particle is perpendicular to the radial line at P. The speed of particle when it is at distance nR, where $n > 1$, from the line charge



A) $\sqrt{V_0^2 + \frac{\lambda Q}{\pi\varepsilon_0 \, m}\ell n 2n}$ B) $\sqrt{V_0^2 + \frac{2\lambda Q}{\pi\varepsilon_0 m}\ell nn}$

C) $\sqrt{V_0^2 + \frac{\lambda Q}{\pi\varepsilon_0 \, m}\ell nn}$ D) none of these

14. A convex lens of focal length 20 cm and another plano-convex lens of focal length 40 cm are placed co-axially. The plane surface of the plano-convex lens is silvered. An object O is kept on the principal axis at a distance of 10 cm from the convex lens. Find the distance d between the two lenses so that the final image is formed on the object itself.



A) 10 cm B) 30 cm C) 20 cm D) 40 cm

15. The radius of hydrogen atom in the ground state is $0.53\mathring{A}$. After collision with an electron, it is found to have a radius of $2.12\mathring{A}$. The principal quantum number ' n ' of the final state of the atom is

A) $n = 3$ B) $n = 4$ C) $n = 1$ D) $n = 2$

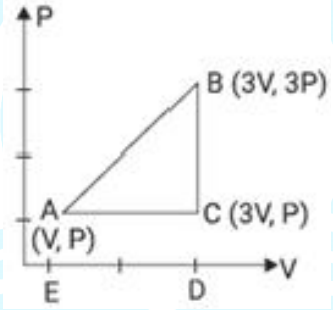
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16. An infinitely long wire has uniform linear charge density $\lambda = 2\text{nC/m}$. The net flux through a Gaussian cube of side length $\sqrt{3}\,\text{cm}$, if the wire passes through any two corners of the cube that are maximally displaced from each other, would be $x\text{Nm}^2\text{C}^{-1}$, where x is:

[Neglect any edge effects and use $\frac{1}{4\pi\epsilon_0} = 9\times10^9\,\text{Nm}^2\text{C}^{-2}$ (SI units)]

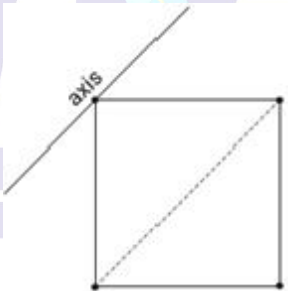
A) 0.72π B) 1.44π C) 6.48π D) 2.16π

17. An ideal gas is taken around $ABC\,A$ as shown in the $P-V$ diagram. The work done during a cycle is



A) $2PV$ B) PV C) $\frac{1}{2}PV$ D) Zero

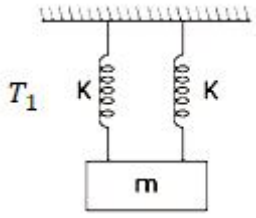
18. Four point masses, each of mass m, are fixed at the corners of a square of side l. The square is rotating with angular frequency ω, about an axis passing through one of the corners of the square and parallel to its diagonal, as shown in the figure. The angular momentum of the square about this axis is:

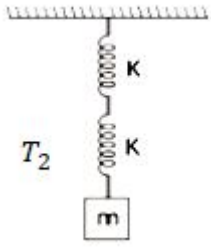


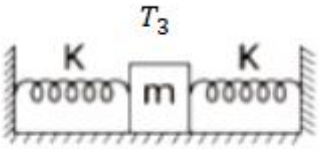
A) $2ml^2\omega$ B) $3ml^2\omega$ C) $ml^2\omega$ D) $4ml^2\omega$

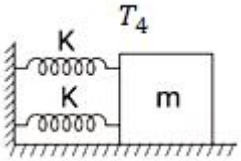
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19. A mass m is attached to two springs of same force constant K , as shown in following four arrangements. If T_1, T_2, T_3 and T_4 respectively be the time periods of oscillation in the following arrangements, in which case time period is maximum?

(a) 

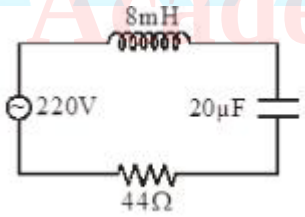
(b) 

(c) 

(d) 

A) (a) B) (b) C) (c) D) (d)

20. For the series LCR circuit shown in the figure, what is the resonance frequency and the amplitude of the current at the resonating frequency.



A) 2500 rad s^{-1} and $5\sqrt{2}$ A B) 2500 rad s^{-1} and 5A

C) 2500 rad s^{-1} and $\frac{5}{\sqrt{2}}$ A D) 25 rad s^{-1} and $5\sqrt{2}$A

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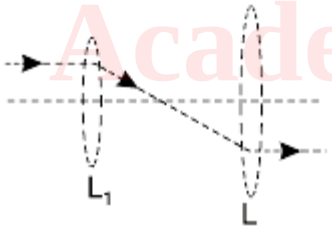
SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

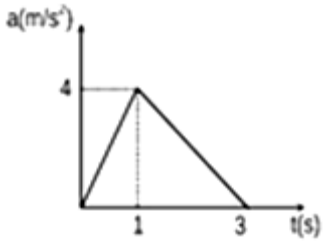
21. A solenoid of length 0.5 m has a radius of 1 cm and is made up of ' m ' number of turns. It carries a current of 5 A. If the magnitude of the magnetic field inside the solenoid is 6.28×10^{-3} T, the value of m is:

22. A heavy nucleus having mass number 200 gets disintegrated into two small fragments of mass numbers 80 and 120. If binding energy per nucleon for parent atom is 6.5 MeV and for daughter nuclei is 7 MeV and 8 MeV, respectively, then the energy released in the decay will be : (in MeV)

23. The following figure represents two biconvex lenses L_1 and L_2 having focal length 10 cm and 15 cm respectively. The distance between L_1 and L_2 (in cm) is:



24. The acceleration time graph of a particle moving in a straight line is as shown in the figure. The velocity of the particle at time $t=0$ is 2 m/s. The velocity after 2 second (in m/s)



25. If the length, breadth and thickness of a strip are (10.0 ± 0.1)cm, (1.00 ± 0.01)cm and (0.100 ± 0.001)cm, respectively, the maximum percentage error in its volume will be

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which of the following statement is not true for the reaction given below?

$$\left[Cu(H_2O)\right]^{2+} + 4NH_3 \rightleftharpoons \left[Cu(NH_3)_4\right]^{2+} + 4H_2O$$

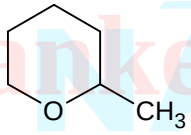
A) It is ligand substitution reaction.

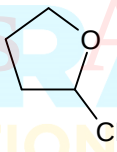
B) NH_3 is relatively strong field ligand while H_2O is weak field ligand

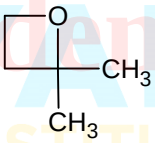
C) During the reaction, there is a change in colour form light blue to dark blue

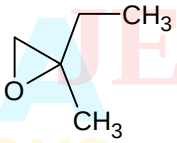
D) $\left[Cu(NH_3)_4\right]^{2+}$ has tetrahedral structure and is paramagnetic

27. A chiral $C_5H_{10}O$ ether reacts with HI to give a $C_5H_{10}I_2$ product. Treatment of this with hot KOH in ethanol produces $1,3-$pentadiene. What is the structure of original ether ?

(A) 

(B) 

(C) 

(D) 

28. 60 % of a first order reaction was completed in 60 min. The time taken for decompose to half of their original amount will be :

A) ≈ 30 min B) ≈ 45 min C) ≈ 20 min D) ≈ 40 min

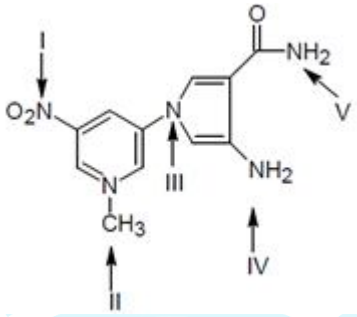
29. The calculated spin-only magnetic moment of $K_3[Fe(CN)_6]$ and $K_4[Fe(CN)_6]$ respectively are :

A) 4.90 and 4.90 BM B) 1.72 and 0.00 BM

C) 4.90 and 5.92 BM D) 5.92 and 4.90 BM

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30. Kjeldahl's method is applicable for the nitrogens as indicated



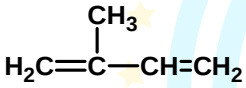
A) All except II and III

B) All except III

C) IV and V only

D) I and IV only

31. Using bond energy data, calculate heat of formation of isoprene



Given bond energy of $C-H=98.8\,kcal$, $H-H=104\,kcal$,

$C-C=83\,kcal, (C=C)=147\,kcal$ and ΔH of $C(s)\rightarrow C(g)=171\,kcal$

A) $28.3\,kcal\,mol^{-1}$

B) $20.6\,kcal\,mol^{-1}$

C) $283\,kcal\,mol^{-1}$

D) $206\,kcal\,mol^{-1}$

32. Mark out the incorrect match with the shape

A) NH_2^- : Pyramidal

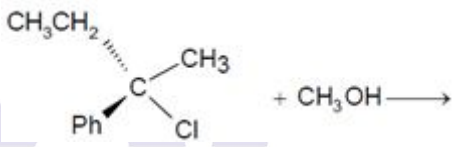
B) ICl_4^- : Square planar

C) $XeOF_2$: T-shape

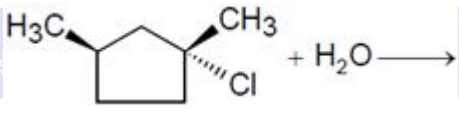
D) XeF_2 : linear

33. A pair of diastereomers is formed in the following reaction

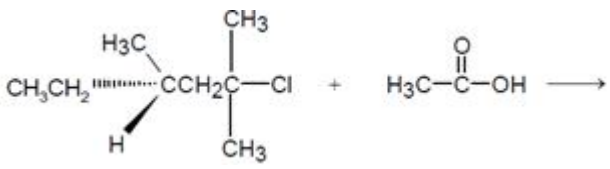
A)



B)



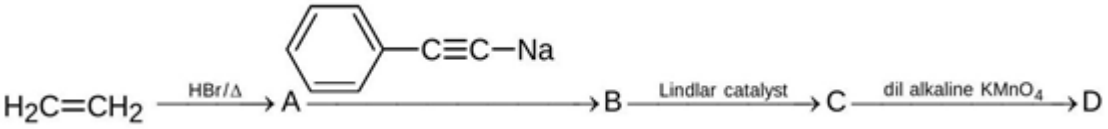
C)



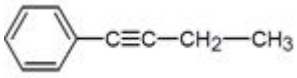
D) All of these

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34. Based on the following sequence of reactions, select **incorrect** statements



A) A is primary alkyl bromide

B) B is 

C) C is a cis-isomer

D) D is obtained by anti addition

35. Given below are two statements:

Statement-I : When mercuric iodide is added to the aqueous solution of KI, the freezing point is raised.

Statement-II : HgI_2 reacts with KI forming complex ion $[HgI_4]^{2-}$.

In the sight of the above statements, choose the correct answer from the options given below:

A) Statement I is true but statement II is false

B) Both statement I and statement II are false

C) Statement I is false but statement II is true

D) Both statement I and statement II are true

36. Statement-1: If an electron is located within the range of $0.1A^0$, then the uncertainty in velocity is approximately 6×10^6 m/s.[h = 6.6×10^{-34}, $m_e = 9.1 \times 10^{-31}$ kg]

Statement - 2 : Trajectory (path of motion) of above electron can be defined.

(A) Statement-1 is True, Statement-2 is True;

(B) Statement-1 is True, Statement-2 is True;

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

37. The order of basic character of given oxides is :

A) Na_2O > MgO > CuO > SiO_2

B) MgO > SiO_2 > CuO > Na2O

C) SiO_2 > MgO > CuO > Na2O

D) CuO > Na_2O > MgO > SiO_2

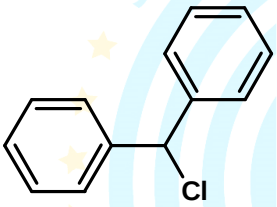
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38. Xenon trioxide (XeO_3) forms xenate ion in alkaline medium. $XeO_3 + NaOH \rightarrow Na[HXeO_4]$. But the xenate ions slowly disproportionate in alkaline solution as $Na[HXeO_4] + NaOH \rightarrow Z + Xe + O_2 + H_2O$.

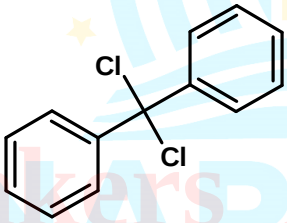
The compound Z is expected to be

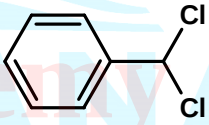
A) Na_2XeO_3 B) Na_2XeO_4 C) Na_4XeO_6 D) Na_4XeO_4

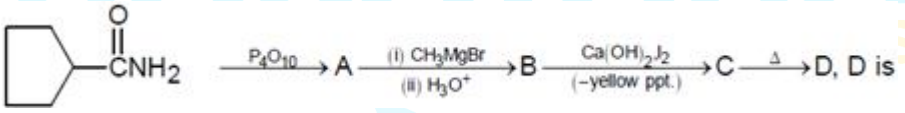
39. Which of the following is obtained as the product when excess benzene reacts with CH_2Cl_2 in the presence of anhydrous $AlCl_3$.

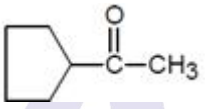
A) 

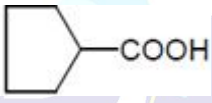
B) 

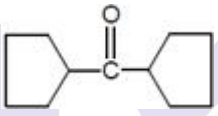
C) 

D) 

40. 

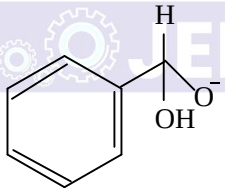
A) 

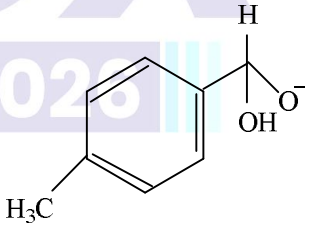
B) 

C) 

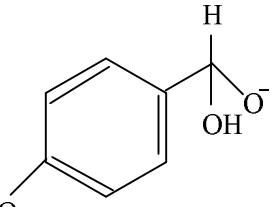
D) 

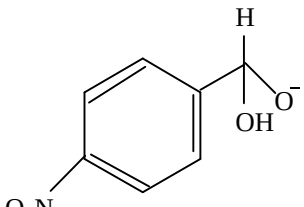
41. In Cannizzaro reaction, the intermediate that will be the best hydride donor is

A) 

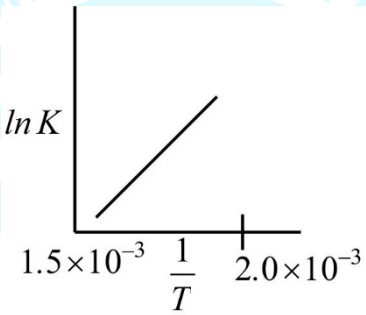
B) 

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C) 

D) 

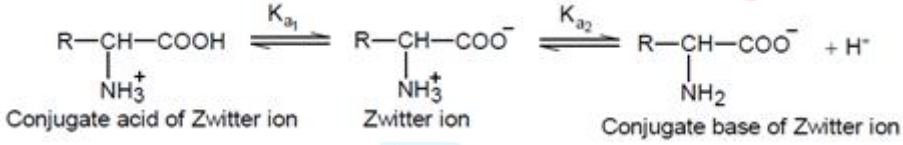
42. The graph relates $\ln K_{eq.}$ vs. $\frac{1}{T}$ for a reaction, the reaction must be



A) exothermic B) endothermic

C) ΔH is negligible D) spontaneous

43. Which statement wrong about $R-\underset{\underset{NH_2}{|}}{CH}-COOH$, if the following equilibrium exist



A) $\left[H^+\right]=\sqrt{K_{a_1}K_{a_2}}$

B) $pH=\frac{pK_{a_1}+pK_{a_2}}{2}$

C) The pH of aqueous solution depends upon concentration $R-\underset{\underset{NH_2}{|}}{CH}-COOH$

D) Zwitter ion behaves as amphoteric salt such as HCO_3^- in aqueous solution

44. Electronegativity values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of apatite family:

A) 3.0 B) 2.1 C)2.0 D)1.9

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45. When a cleaned strip of zinc metal is placed in a solution of $CuSO_4$, a spontaneous reaction occurs. And following observation(s) is/are made?

(a) the mass of zinc metal decreases gradually

(b) the copper metal starts depositing on either zinc plate or settles down to the vessel

(c) the solution remains electrically neutral

(d) the temperature of the solution decreases as it is an endothermic reaction.

Choose the correct answer from the options given below:

A) a, b and c only B) a, c and d only

C) a and c only D) b, c and d only

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. A bond X–Y has a dipole moment of 1.8×10^{-29} Cm and a bond length 150 pm. What will be the percentage of ionic character in X–Y bond.

47. There are some species given below.

(i) O_2^+ (ii) CO (iii) B_2 (iv) O_2^-

(v) NO^+ (vi) He_2^+ (vii) C_2^{2+} (viii) CN^-

(ix) N_2

Total number of species which have their fractional bond order.

48. 1 mole of phenol and 1 mole of benzoic acid on reaction with $NaHCO_3$ at 300 K and 1 bar forms ….Xdm^3CO_2. Thus, the value of X is_____

49. In the Claisen – Schmidt reaction 1 mole of benzaldehyde and 1 mole of acetaldehyde are involving to form 92.4 grams of cinnamaldehyde. The percentage yield in this reaction was…….%

The reaction is

$$Ph-CHO+CH_3-CHO \xrightarrow{OH^-} Ph-CH=CH-CHO$$

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50. K_a for HCN is 5×10^{-10} at $25^\circ C$. For maintaining a constant pH of 9, the volume of 5 M KCN solution required to be added to 10 mL of 2 M HCN solution is____

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Suppose $\alpha \neq \frac{k\pi}{2}$ for any integer k and $T = \frac{\sin\alpha+\tan\alpha}{\cos\alpha+\cot\alpha}$. Which of the following statements is true?

A) T takes negative values only

B) T takes non-negative values only

C) T takes positive values only

D) T takes both positive and negative values

52. If a, b and c are consecutive positive integers and $\log(1+ac) = 2k$, then the value of k is

A) log a B) log b C) log c D) log abc

53. Exhaustive set of domain of definition of the function $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$, is:

A) (1,2) B) $(-1,0) \cup (1,2)$

C) $(1,2) \cup (2, \infty)$ D) $(-1,0) \cup (1,2) \cup (2, \infty)$

54. There are two urns. There are m white & n black balls in the first urn and p white & q black balls in the second urn. One ball is taken from the first urn & placed into the second. The probability of drawing a white ball from the second urn is

A) $\frac{(p+1)n+pm}{(m+n)(p+q+1)}$ B) $\frac{(p+1)m+pn}{(m+n)(p-q+1)}$

C) $\frac{(p+1)m+2pn}{(m+n)(p+q+1)}$ D) $\frac{(p+1)m+pn}{(m+n)(p+q+1)}$

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55. Let $G(x)=\left(\frac{1}{a^x-1}+\frac{1}{2}\right)F(x)$, where a is a positive real number not equal to 1 and $F(x)$ is an odd function. Which of the following statements is true?

A) $G(x)$ is an odd function

B) $G(x)$ is an even function

C) $G(x)$ is neither an odd function nor an even function

D) Whether $G(x)$ is an odd or even function depends on the value of a

56. Let $a>0$. If the function $f(x)=6x^3-45ax^2+108a^2x+1$ attains its local maximum and minimum values at the points x_1 and x_2 respectively such that $x_1x_2=54$, then $a+x_1+x_2$ is equal to

A) 15 B) 18 C) 24 D) 21

57. Let $f(x)=\int x^3\sqrt{3-x^2}dx$. If $5f\left(\sqrt{2}\right)=-4$, then $f(1)$ is equal to

A) $-\frac{2\sqrt{2}}{5}$ B) $-\frac{8\sqrt{2}}{5}$ C) $-\frac{4\sqrt{2}}{5}$ D) $-\frac{6\sqrt{2}}{5}$

58. If the letters of the word MATHS are permuted and all possible words so formed are arranged as in a dictionary with serial numbers, then the serial number of the word THAMS is _____

A) 103 B) 102 C) 101 D) 104

59. If $\tan\alpha+\cot\alpha=a$ then the value of $\tan^4\alpha+\cot^4\alpha=$

A) a^4+4a^2+2 B) a^4-4a^2+2 C) a^4-4a^2-2 D) a^4-2a^2+2

60. If 5^{97} is divided by 52, then the remainder obtained is

A) 3 B) 5 C) 4 D) 0

61. Consider the function $f(x)=\frac{3x-1}{x^2+x}$. Which of the following is not correct?

A) $f(x)$ is one-one $\forall x\in(1,\infty)$

B) range of $y=f(x)$ is $(0,1)$ if $x\in(1,\infty)$

C) range of $y=f(x)$ is $[9,\infty)$ if $x\in(-1,0)$

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D) $f(x)$ is one-one $\forall x \in (-1,0)$

62. If $f(x) = 5^{g(x)}$ and $g(x) = \int_2^{x^2} \frac{t}{\ell n\ (1+t^2)}\ dt$, then find the value of $f'(\sqrt{2})$.

A) $4\sqrt{2}$ B) 4 C) 2 D) 16

63. If 2 data sets having 10 and 20 observations have coefficients of variation 50 and 60 respectively and arithmetic means 30 and 25 respectively, then the combined variance of those 30 observation is

A) $\frac{2075}{3}$ B) $\frac{2075}{9}$ C) $\frac{1000}{9}$ D) $\frac{1075}{3}$

64. The number of skew symmetric matrices of order 4×4, where entries belong to the set $\{-3,-2,-1,0,1,2,3\}$ is

A) 5^9 B) 7^9 C) 5^6 D) 7^6

65. If $f(x) = \begin{cases} \frac{x^2-1}{x^2+1} & , \ 0 < x \le 2 \\ \frac{1}{4}(x^3 - x^2) & , \ 2 < x \le 3 \\ \frac{9}{4}(|x-4| + |2-x|) & , \ 3 < x < 4 \end{cases}$, then:

A) f (x) is differentiable at x = 2 & x = 3

B) f (x) is non-differentiable at x = 2 & x = 3

C) f (x) is differentiable at x=3 but not at x=2

D) f (x) is differentiable at x = 2 but not at x = 3.

66. The system of equations $\begin{pmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b \\ 3 \\ -1 \end{pmatrix}$ has no solution if a and b are

A) $a = -3,\ b \neq 1/3$ B) $a = 2/3,\ b \neq 1/3$

C) $a \neq 1/4,\ b = 1/3$ D) $a \neq -3, b \neq 1/3$

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67. Let $\lim\limits_{x\to\pi}\dfrac{\sin\alpha x}{1-2\cos\beta x}=-\sqrt{3},\alpha\in I^{+},\beta>0$, then the least positive value of $\alpha\beta$ is

A) $\frac{1}{6}$ B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) 1

68. Consider $f(x)=x^3+ax^2+bx+c$. Parameters a,b and c are chosen respectively by throwing a dice three times. Then the probability that $f(x)$ is an increasing function is

A) $\frac{5}{36}$ B) $\frac{8}{36}$ C) $\frac{1}{3}$ D) $\frac{4}{9}$

69. Let α and β are the eccentric angles of the end points of the focal chord of the hyperbola $\frac{x^2}{9}-\frac{y^2}{7}=1$, then the possible value of $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}$ is

A) – 7 B) 7 C) $\frac{1}{5}$ D) – 5

70. The length of the perpendicular drawn from the point (3, –1, 11) to the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is:

A) $\sqrt{29}$ B) $\sqrt{33}$ C) $\sqrt{53}$ D) $\sqrt{66}$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. The equation of chord of ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ whose mid point is (3, 1) is ax+by=169, then a+b is equal to

72. Let the roots of the equation $x^4-4x^3+6x^2-4x-15=0$ are $\alpha,\beta,\gamma,\delta$, where $\alpha,\beta\in R$ then the value of $\gamma\delta-\alpha\beta$ is

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73. If the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x}{1}=\frac{y}{\alpha}=\frac{z-5}{1}$ is $\frac{5}{\sqrt{6}}$, then the absolute value of sum of all possible values of α is

74. The sum of the series, $\cot^{-1}\left(1+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+\ldots=\infty=\tan^{-1}(k)$, then the value of k is

75. If $\alpha,\beta\neq 0$ and $f(n)=\alpha^n+\beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K\,(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$, then K is equal to

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